Weekly Report 1: Web-Based Real-Time Multiplayer Ludo Game

Date: July 13, 2026

Team Members:

- **Shihab Uddin Patwary** (ID: 2132685642)
- **Md. Sajidul Islam** (ID: 2121935642)
- **Mohiuddin Sefat** (ID: 2231193642)
- **Sakiha Zaman** (ID: 2233351642)
- **Ishraq Hossain Rodro** (ID: 2311675042)
- **Musanna Ibn Amin** (ID: 2233285042)

This week, we laid the foundation for the real-time multiplayer Ludo platform. The system is split into independent micro-modules, establishing a running local server environment, live client-server socket channels, baseline database schemas, and the foundational geometric array definition for the 15x15 game layout.

Detailed Task Accomplishments

1. Frontend Wireframe & UI Layout

- **Boilerplate Setup:** Initialized the repository structure and frontend configurations utilizing responsive web paradigms.
- **Static Layouts Completed:** Designed and deployed public/index.html (Welcome interface), public/login.html (Authentication screen), and public/signup.html (Registration interface).
- **Component Integrations:** Programmed an interactive, togglable password-reset modal workflow within the login framework and configured client-side Socket scripts to auto-handshake with the running server interface upon page load.

2. Database Layer & Schema Construction

- **Database Infrastructure:** Configured config/db.js to manage an asynchronous Mongoose wrapper connecting securely to the local MongoDB engine environment.
- **Data Object Blueprints:** Engineered models/User.js containing precise constraints for persistent player profile accounts.
- **Guest Track Models:** Established schema wrappers isolated explicitly for temporary, non-authenticated guest account sessions to track volatile leaderboard standings during active play.

3. Duplex Real-Time Networking Prototype

- **Network Initialization:** Built a standalone communication layer inside network/socketHandler.js configured with standard CORS parameters to facilitate cross-origin network requests.
- **Connection Events Lifecycle:** Developed foundational handlers to capture live client connection connections and trace abrupt disconnect triggers.
- **Data Payload Sandbox:** Implemented a verification pipeline loop (ping_test / pong_test) allowing the UI to send data parameters and receive immediate receipts, confirming operational duplex pathways.

4. Mathematical Game Matrix Formulation

- **Map Geometry Definitions:** Generated the comprehensive structural array definitions inside ludoBoardMatrix.js.
- **Cell Segment Coordinates:** Successfully indexed the 52 global common tracks, standard safety tiles, and localized player home-run destination pathways.
- **Data Exposing API:** Linked the structure array configuration directly to the backend web core, introducing a verification route (/api/map) to let future frontend elements read the grid geometry instantly.

Next Week's Objectives

1. Bind the mathematical data track arrays directly to a responsive 15x15 CSS Grid / Tailwind canvas board layout.
2. Shift the game loop rules (dice outcomes, boundary restrictions, piece captures) safely onto server-authoritative backend listeners.
3. Deploy room routing structures to cleanly group up to 4 concurrent socket streams into isolated private match instances.

Report compiled and submitted for review by the project team members listed above.